

Nikolay Malkovsky

Principal Researcher | Scientific computing | High-Performance C++

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WORK EXPERIENCE

Principal Researcher

2023–Present

Huawei | 2012 Laboratories

- Improved QUIC for file transfer and real-time communication using congestion-control algorithms.
- Technical lead for R&D on forward error correction in real-time data transfer, including advanced Reed–Solomon and convolutional-code implementations.
- Authored the [TechArena 2025 SPbSU challenge \(codeforces.com/gym/106111\)](https://codeforces.com/gym/106111).

Developer

2022–2023

Yandex | Yandex Market

Senior Researcher

2021–2022

TCS Group | AI Department

- Built transformer-based zero-shot text cleaning.
- Developed core components for CTC end-to-end ASR and WFST-based context biasing.

Researcher / Senior Researcher

2018–2021

STC Innovations | Research Department

- Developed the HMM component of hybrid HMM–DNN ASR: finite-automata word-boundary extraction, low-latency streaming recognition, vocabulary expansion, lattice generation, confusion-network conversion, and RNN-LM lattice rescoring.

Assistant Professor

2017–2022

St. Petersburg State University | and HSE University, St. Petersburg campus

- Taught a Russian-language course on [convex optimization \(course materials\)](#).

OPEN SOURCE

Author and Maintainer

2025–Present

Pixie | C++20 succinct data structures library

- Highly optimized compressed data structures with fast access without full decompression, focused on practical memory efficiency and CPU performance.
- github.com/Malkovsky/pixie

Author and Maintainer

2026–Present

AI for C++ | Tools for AI usage in HPC and algorithmic tasks mainly for C++

- Recommended MCPs, skills, general guidance for HPC workflows
- github.com/Malkovsky/ai_for_cpp

Author

2018–2024

Interactive visualization | Python library for interactive widgets

- Set of tools for creating interactive visualization in Python mainly designed for educational purposes
- github.com/Malkovsky/interactive-visualization/

JOB PREFERENCES

I specialize in scientific computing and have consistently performed best on projects that are mathematically demanding and require high-performance algorithms—areas that also align closely with my personal interests. I am particularly interested in R&D projects involving technically challenging computational problems, such as neural network inference, algebraic computation, and large-scale data processing.

EDUCATION

Ph.D. in Computer Science

2013–2017

St. Petersburg State University
St. Petersburg, Russia

Thesis: “*Randomized resource distribution algorithms in multiagent systems.*”

Supervisor: **Prof. Oleg Granichin**.

Candidate of Science (Candidate of Physical and Mathematical Sciences in the Russian degree system).

Master's Degree

2008–2013

St. Petersburg State University
St. Petersburg, Russia

GPA: 4.5/5

LANGUAGES

Russian

Native

English

Fluent

SKILLS

Algorithms & systems

As a techlead developed performance-critical real-time systems which demanded advanced algorithms: graph and finite-automata algorithms; finite fields and coding theory; succinct data structures.

C / C++

Primary languages; leadership and maintenance of algorithmically rich projects of medium scale (10^5 – 10^6 lines of code)

AI adoption

I use AI a lot in my workflow including HPC and algorithmic tasks, constantly seek for ways to improve efficiency of AI tools

Python

Main scripting language: NumPy, SciPy, scikit-learn, Matplotlib, PyTorch, and TorchScript; C/C++ extensions for bottlenecks.

Mathematics

Strong mathematical background applied throughout major research and engineering projects.

Speech recognition

Kaldi, DeepSpeech, NVIDIA NeMo; HMM–DNN, CTC, WFST, context biasing.

Infrastructure

Docker, gRPC, Kubernetes; microservice development.